

Sutariya Tirth Vijaykumar

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Professional Summary

Mathematics and Computing undergraduate at SVNIT with experience in Python, backend development, data-intensive applications, cloud deployment, and scalable software systems. Interested in software engineering, automation, and distributed infrastructure.

Work Experience

GWOC – MindSettler, SVNIT

github.com/InspireEmber/MindSettler-GWOC-26

Dec 2025 – Jan 2025

Software Development Intern

- Developed and deployed a psycho-education platform using Next.js, Node.js, and MongoDB with automated session management.
- Built an AI chatbot using OpenRouter with safety guardrails and implemented OAuth 2.0 authentication workflows.
- Integrated Google Calendar and Brevo Email APIs to automate bookings, reducing manual scheduling effort by over 60%.
- Collaborated on deployment and cloud hosting workflows using Vercel and Render with secure access control.

Projects

Real Estate Investment Analyzer

github.com/Nava-deep-18/Real_Estate_analyser

Dec 2025 – Jan 2025

Python, Streamlit, Pandas, ChromaDB, Gemini LLM, SQLite

- Designed a data processing pipeline routing queries between SQL databases and vector embeddings for investment analysis.
- Built a financial simulation engine for 20-year Buy vs Rent analysis with loan amortization and India-specific tax modeling.
- Developed an ETL pipeline to scrape, clean, impute missing values, and vectorize over 5,000+ property listings.
- Created an analytics dashboard with chat-based querying and visual insights for property market trends.

StochOpt: Deep Stochastic Optimization Engine

github.com/adityamedpalliwar/StochOpt-Engine

Jan 2026 – Feb 2026

Python, PyTorch, Stable-Baselines3, PPO, W&B, DevOps Automation

- Contributed to a stochastic optimization platform for dynamic pricing and inventory management.
- Built and optimized reinforcement learning environment pipelines for PPO-based agent training and evaluation.
- Developed automation workflows for model downloading, artifact synchronization, and evaluation setup using W&B Artifacts.

Skills

Languages: Python, C++, SQL

Backend & Databases: FastAPI, REST APIs, Authentication, PostgreSQL, MongoDB

Data & Libraries: Pandas, NumPy, PyTorch, ChromaDB, Stable-Baselines3, Gemini LLM, OpenRouter

Tools: Git, GitHub, Docker, SQLite, Streamlit, Vercel, Render, Weights & Biases

Concepts: Data Structures, OOP, ETL Pipelines, REST APIs, RAG Systems, Machine Learning

Education

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat

B.Tech in Mathematics and Computing — CGPA: 8.45 — Third Year

Aug 2024 – Present

Soft Skills

Leadership, Communication, Problem Solving, Analytical Thinking, Team Collaboration, Ownership, Adaptability